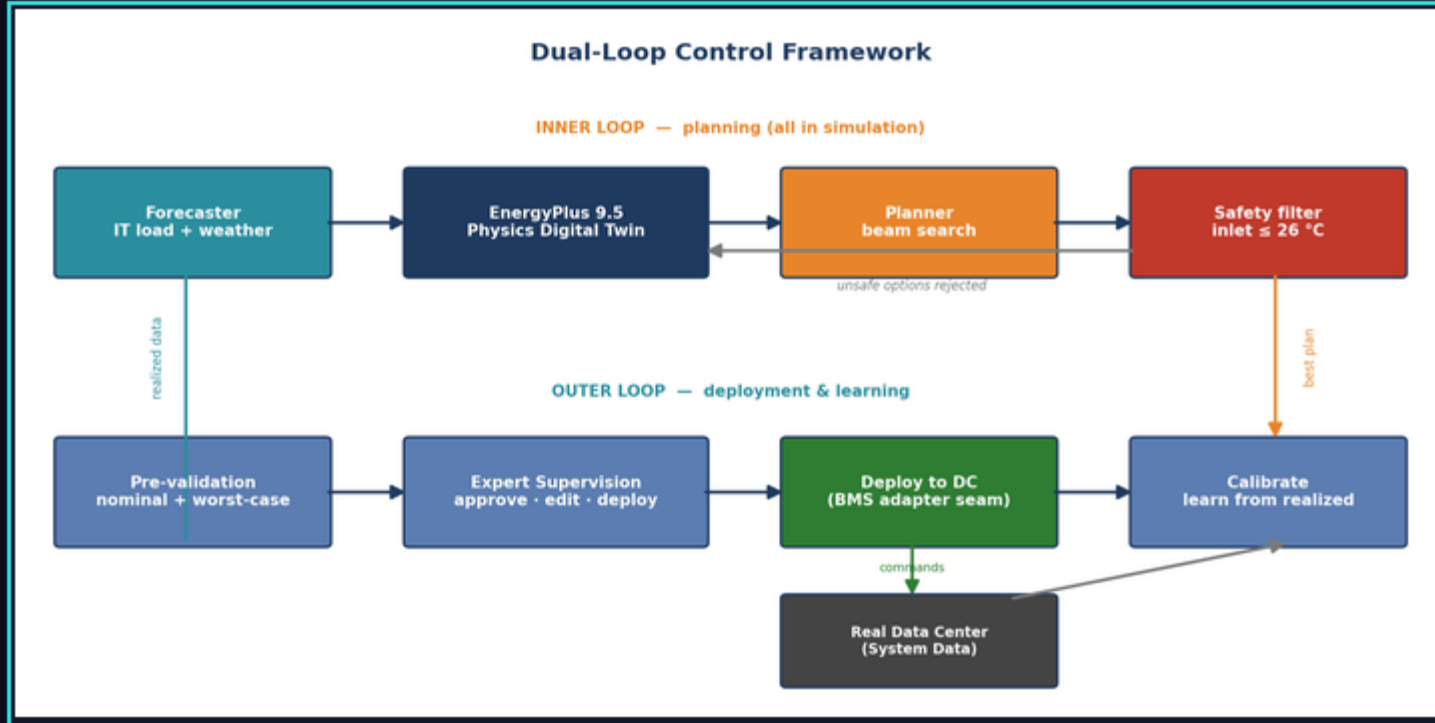

Cheaper Without Crossing the Line

A Digital-Twin Dual-Loop Controller for Data-Center Cooling, Walked Through Its Own Cockpit (EnergyPlus 9.5 in-the-loop; 6 deployed weeks; 0 safety violations)

By making the 26 C inlet cap a hard, non-negotiable constraint and energy a soft objective, a physics-grounded dual-loop controller can run a 2 MW hall warmer and slower for a measured 3.63% weekly HVAC saving while a learning loop drives its prediction error from 1.33% to 0.10% over six weeks and never once touches the cap.

ORIENTATION

One controller, two loops — and a cockpit tab for each part



Read the diagram as two nested loops:

1 INNER loop (simulation): forecast -> EnergyPlus physics -> beam search -> safety filter

2 OUTER loop (deploy & learn): pre-validate -> expert approve -> deploy -> calibrate

Tabs map onto the loops:

3 New Plan + Review = inner loop (plan & prove)

4 Dashboard + Live = the operator's contract & live audit

5 History + 3D = the outer loop closing and made tangible

ACT 0 — THE TRUST BOUNDARY

Governance before physics

Nothing is reachable until a valid, role-separated token is verified — fail-closed.

The door: token-gated, role-separated, fail-closed

• DCTWIN

Enter your access token

Continue

operator or expert token

1 2 3

The image shows a login interface for 'DCTWIN'. It features a text input field for an 'access token', a 'Continue' button that is disabled (indicated by a greyed-out appearance and a red circle with a '1' pointing to it), and a dropdown menu for selecting a role ('operator or expert token', indicated by a red circle with a '2'). A red circle with a '3' points to the bottom of the form area.

How to read it:

1 One masked token field — no user/password, credentials are pre-issued

2 'Continue' is disabled until valid; shows 'Checking...' during a live backend probe

3 Two roles: operator (view + run) vs expert (approve + deploy)

Money insight:

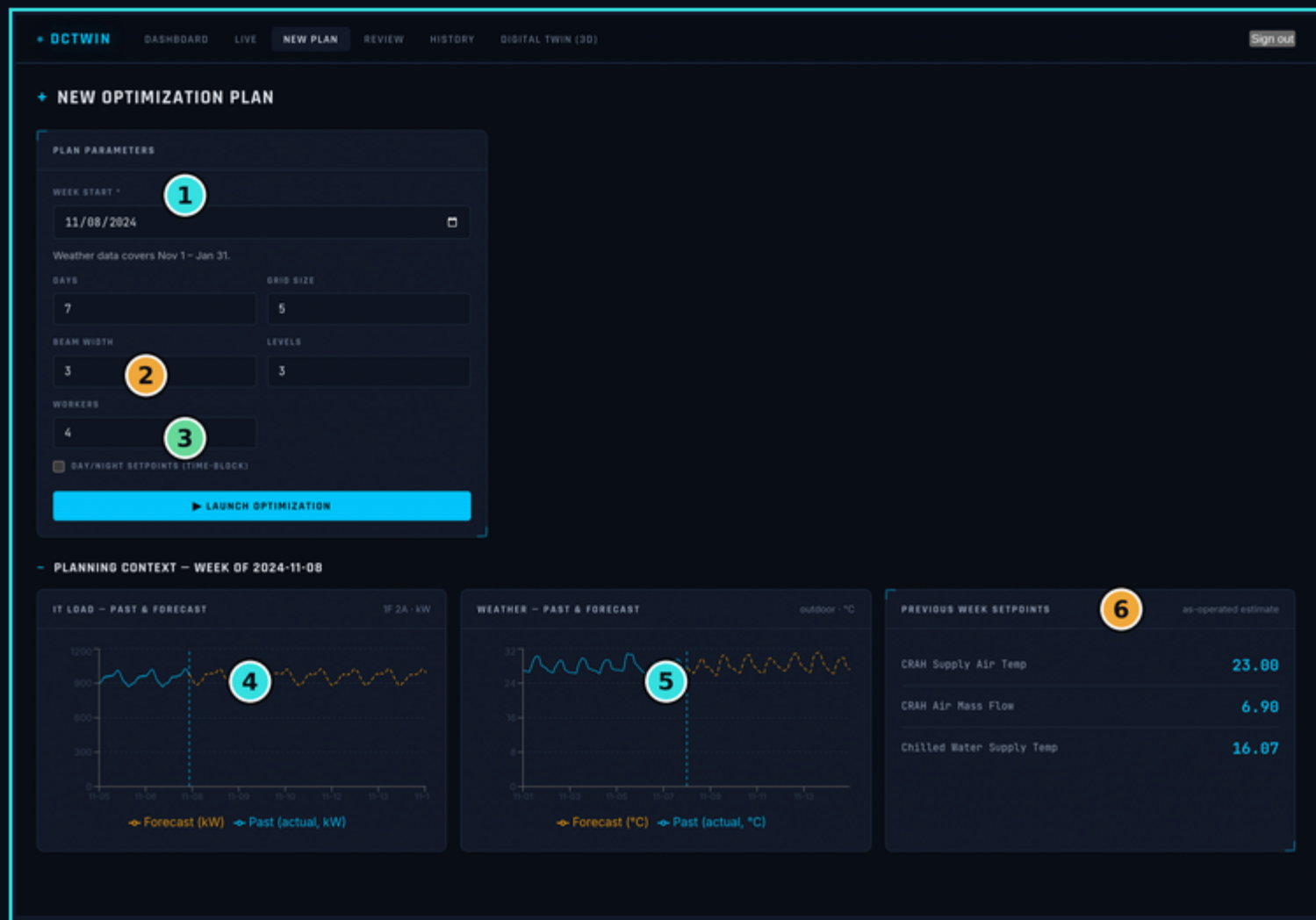
A bad token is never cached; even a backend outage denies entry (fail-closed).

ACT 1 — THE INNER LOOP AS AN EXPERIMENT

45 actuators reduced to a 3-knob weekly search

Each candidate scored by a full-week EnergyPlus 9.5 run; coarse-to-fine beam search.

Launching the week — and showing its inputs



Plan parameters (left):

1 Week, days, grid=5, beam=3, levels=3, workers — the search budget

2 Optional day/night setpoints (time-block search)

3 Launch → hundreds of full-week EnergyPlus runs

Planning context (why this plan):

4 IT-load past + forecast

5 Weather past + forecast (+1 σ hot week hedges safety)

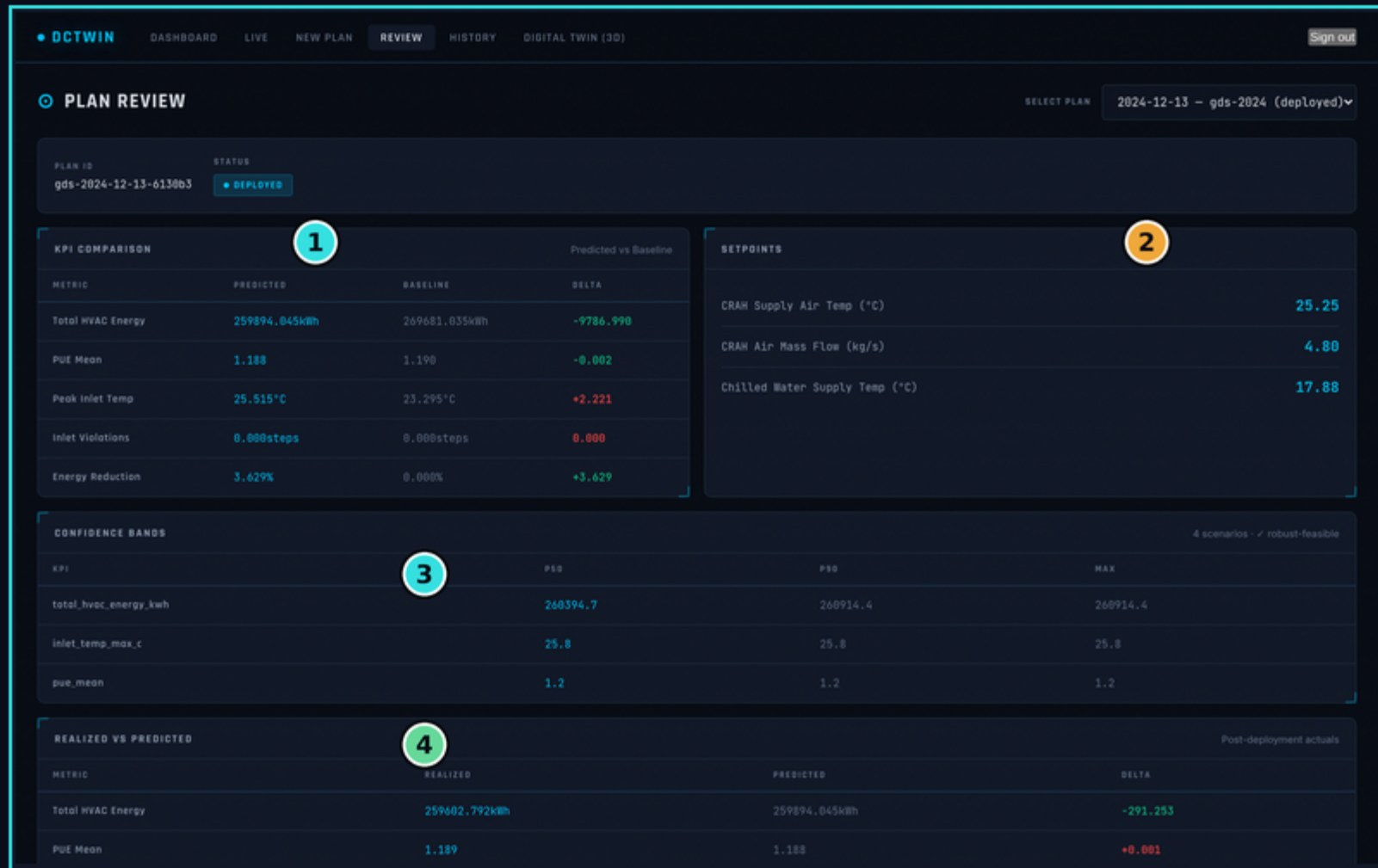
6 Previous-week setpoints (the as-operated baseline)

ACT 2 — SAFETY IS A HARD CONSTRAINT, SAVINGS ARE THE CONSEQUENCE

The evidence gate

Predicted vs baseline, robust confidence bands, and the inlet trajectory against the 26 °C line.

Did it beat the baseline — and did reality agree?



1 KPI comparison: predicted vs baseline (energy, PUE, peak inlet, violations, reduction)

2 Baseline = the operator's own as-operated setpoints

3 Confidence bands p50 / p90 / max from the robust ensemble

4 Realized vs predicted — the twin checked against the deployed week

Reduction here $\approx$ 3.6% with 0 predicted violations.

The safety money-shot: every step stays under the cap



1 KPI bar chart: plan vs baseline at a glance

2 The red 26 °C cap line — the hard, non-negotiable limit

3 Inlet trajectory (nominal + worst-scenario) hugs, but never crosses, the line

This is what 'cheaper without crossing the line' looks like.

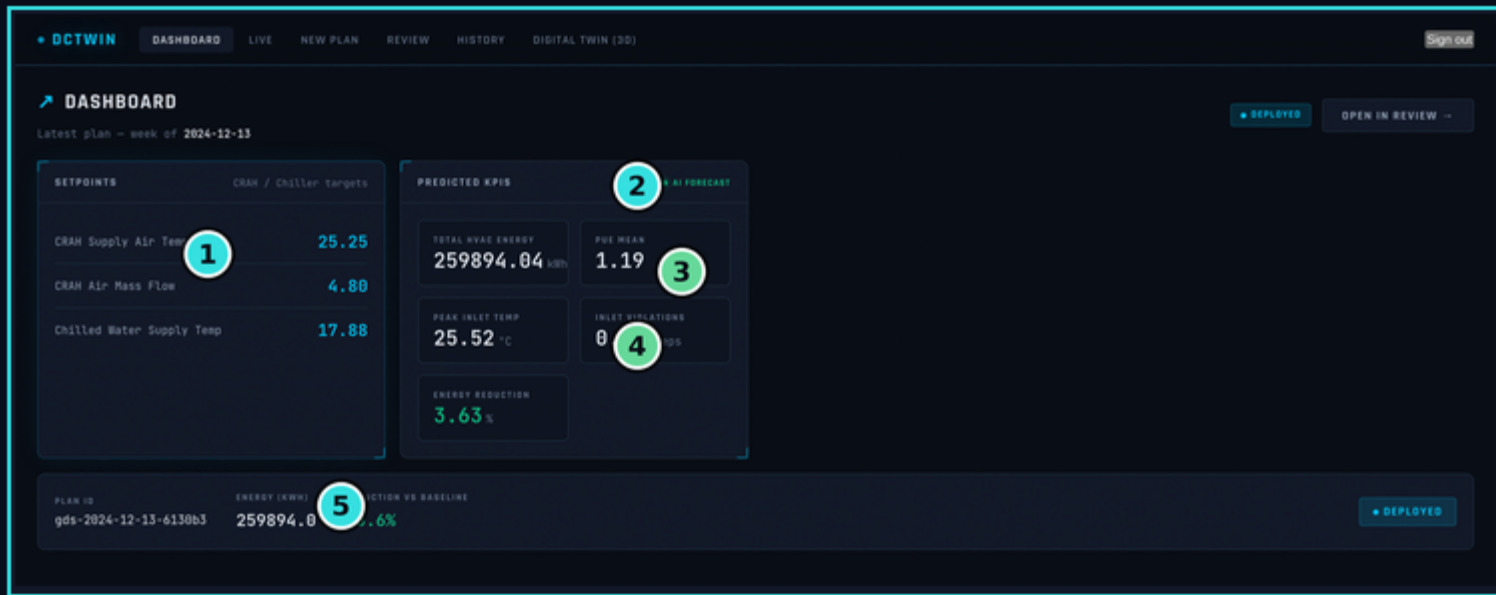
Expert approves/deploys ONLY from this evidence; blocked_unsafe is not approvable.

ACT 3 — THE OPERATOR'S CONTRACT & THE LIVE AUDIT

At-a-glance verdict, then real-time proof

Dashboard = what we're running; Live = the cap is holding right now.

The one-screen answer: safe and cheaper



1 Three active setpoints (SAT 25.25, airflow 4.80, CHWST 17.88)

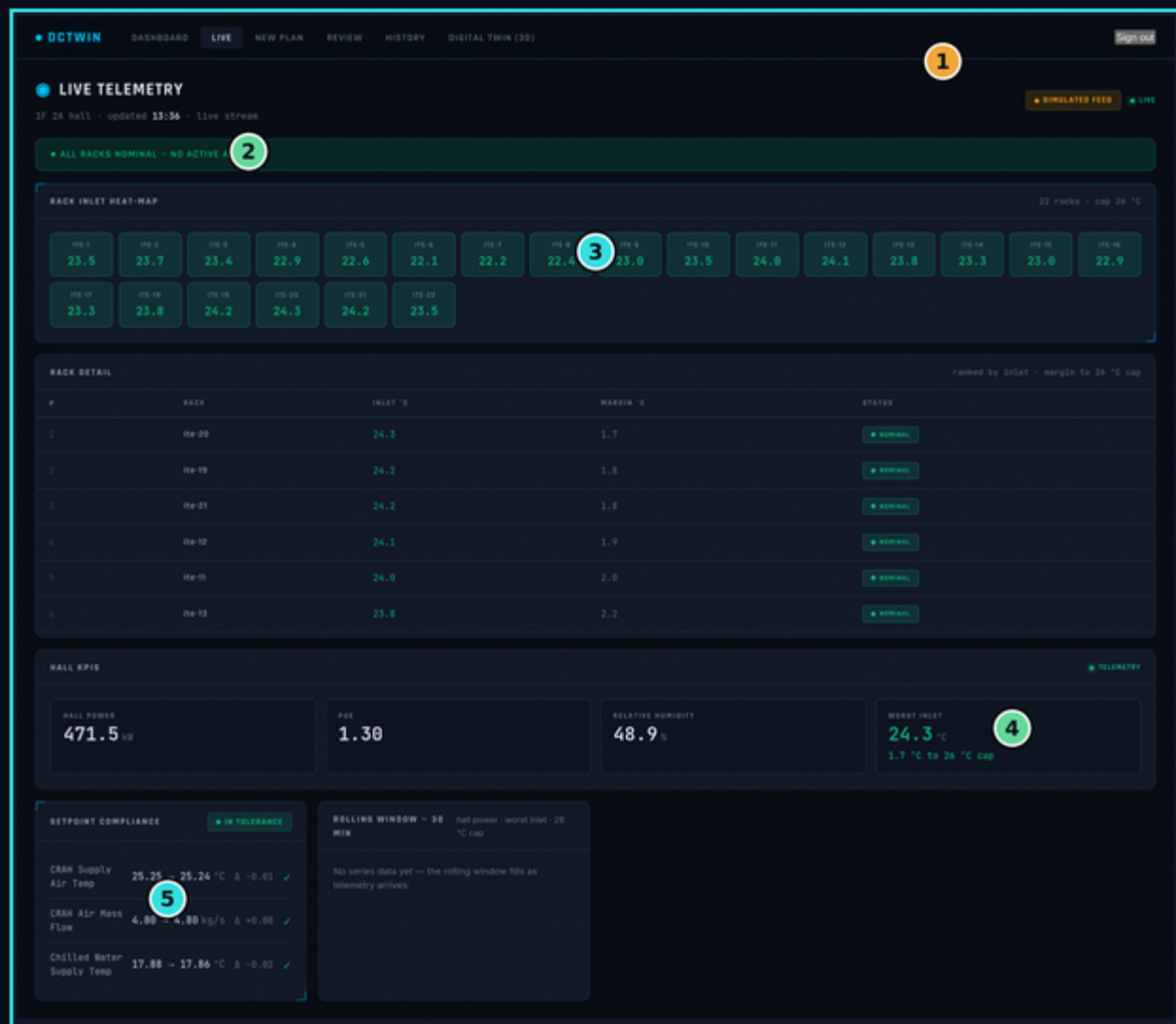
2 Predicted KPIs: energy 259,894 kWh, PUE 1.19

3 Peak inlet 25.52 °C < 26 °C cap; 0 violations

4 Energy reduction 3.63% vs baseline

5 Plan status = DEPLOYED (shadow mode)

Real-time audit: the cap is holding now (Expert Supervision)



1 SIMULATED FEED badge — honesty: this is a labelled sim feed

2 Alert banner: all racks normal (warn $\geq 25^{\circ}\text{C}$, critical $\geq 26^{\circ}\text{C}$)

3 22-rack inlet heat-map — spatial health at a glance

4 Worst inlet 24.3°C $\rightarrow$ 1.7°C of headroom to the cap

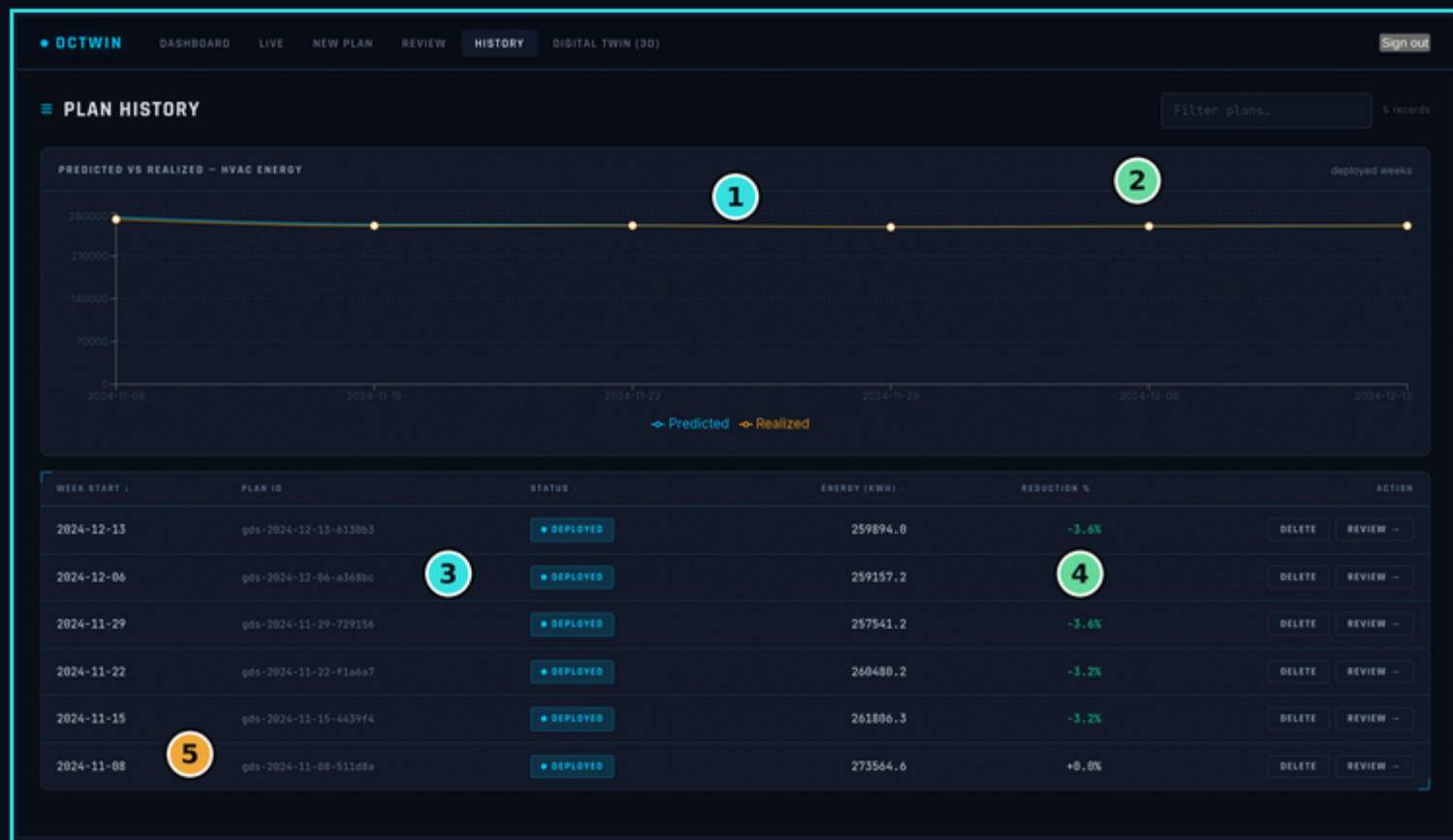
5 Setpoint compliance: commanded vs held, in tolerance

ACT 4 — THE OUTER LOOP CLOSING

Six weeks of convergence, gate-released savings

Prediction error 1.33% → 0.10%; savings 0 → 3.63%; zero violations.

Is the twin learning? Predicted vs realized, week over week



1 Predicted (teal) vs realized (orange) HVAC energy

2 Lines converge as calibration learns → error 1.33% → 0.10%

3 Plan ledger: every deployed week, energy, reduction

4 Reduction ramps 0% → 3.2% → 3.6% (released only as proven safe)

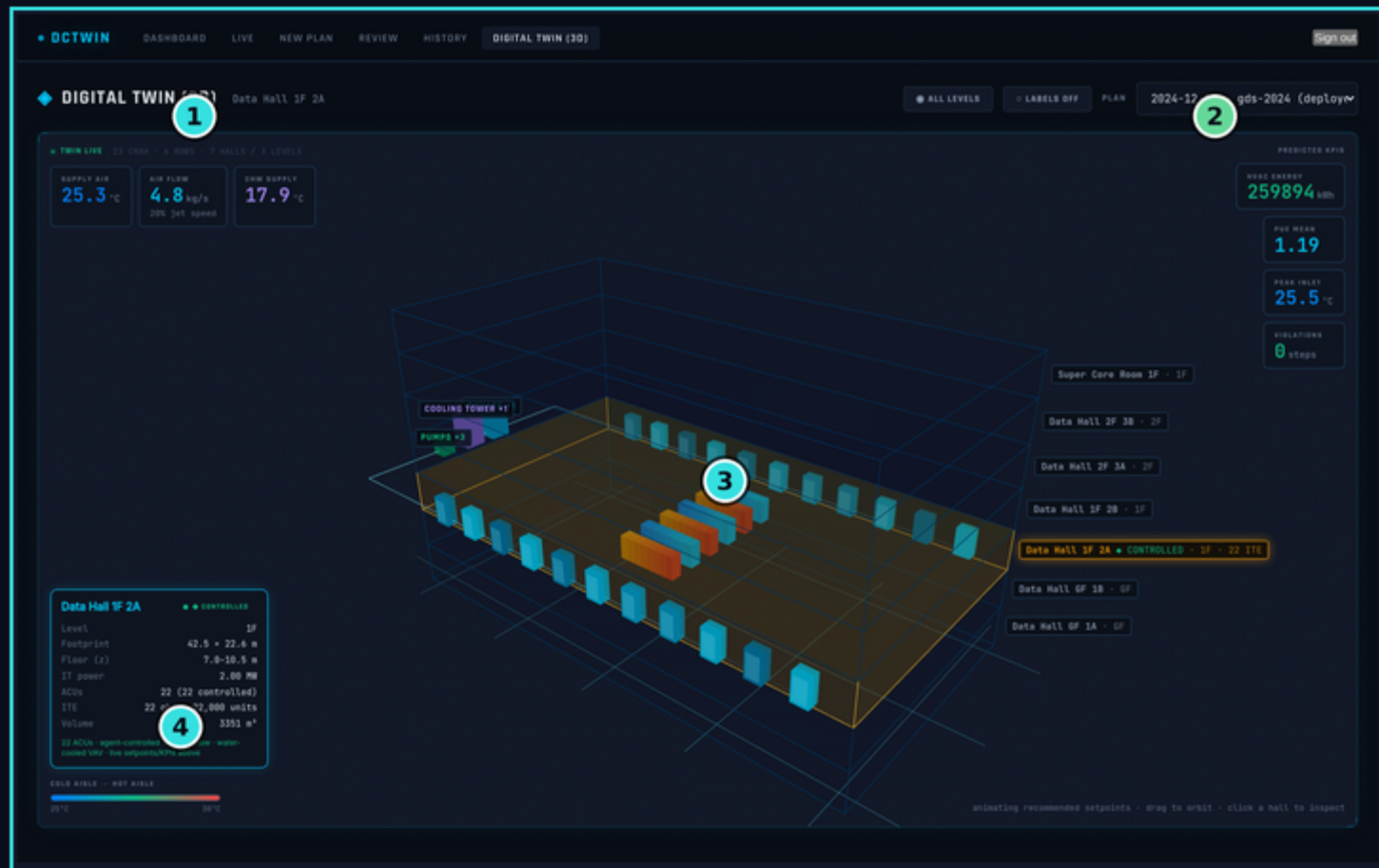
5 Week 1 = 0% (cold start): the gate allowed only the baseline

ACT 5 — THE TWIN MADE TANGIBLE

Seeing the energy-vs-safety trade in space

The controlled hall, rack rows colored live from telemetry.

The physics model you can walk around



1 HUD: active plan setpoints (SAT / airflow / CHWST)

2 HUD: live KPIs (energy, PUE, peak inlet, violations)

3 Rack rows colored from live telemetry — hotspots are visible in space

4 Color legend = inlet temperature scale toward the 26 °C cap

ACT 6 — HONEST LIMITS

Structurally proven, not yet field-proven

And the single remaining seam to a live building.

What is real, what is simulated, what remains

Real and exercised today

- Full-physics EnergyPlus 9.5 oracle, the 3-net safety gate, the calibration/learning loop, the whole cockpit
- 6 deployed weekly cycles, all tests green; numbers shown are read from the live run database

Deliberately simulated / shadowed (and labelled)

- Deploy = SHADOW MODE: 45 commands recorded, never actuated
- Telemetry feed is SIMULATED (the SIMULATED FEED badge); 'realized' = perturbed-plant EnergyPlus stand-in

The one remaining seam to the field

- Connect a real historian to POST /api/telemetry + implement the BACnet adapter behind the existing stub
 - Then: recommend-only for 4-8 weeks, calibrate recirculation from real rack sensors, then actuate hall-by-hall

The hardest cross-cutting questions

- You deliberately run the hall ~ 2.2 C hotter than the operators do (25.5 C vs 23.3 C peak inlet)....
- Your predicted and realized energy lines are nearly identical, and 'realized' comes from EnergyPlus...
- Beam search over three setpoints feels crude. Why not Bayesian optimization or gradients, and how...
- You keep quoting two PUE numbers, 1.19 and 1.30. Which is real, and are you cherry-picking the...

...and the rest of the hard ones

- Six weeks on one hall, with a flat IT load — is this statistically meaningful or a toy?
- On the Review inlet-trajectory chart, the worst-case red line clearly pokes above 26 C at the very...
- You call deploy 'shadow mode' — so has this system ever actually controlled a data center?

CLOSING

Cheaper Without Crossing the Line

Hard safety constraint · soft energy objective · a learning loop — measured over six weeks.

The measured record — six deployed weeks

Accuracy

- Weekly-energy prediction error 1.33% → 0.10%; peak-inlet error within ~0.25 °C

Savings

- 0% → 3.2% → 3.63% vs the as-operated baseline, released as confidence grew

Safety

- Zero violations; realized peak inlet rose 23.37 → 25.60 °C, never reached the 26 °C cap (0.40 °C margin)

Well-posedness (real E+ sweep)

- Airflow moves hall energy 15.5%, CHWST 3.4%, SAT ~0.5%

Calibration state (n=7)

- inlet bias 0.18 °C, σ_{inlet} 0.14 °C, σ_{post} 0.36 °C, energy bias ~0.2%

Delivery tips (gist on slide, full text in notes)

- Lead every claim with safety, then reveal savings as the consequence — never the reverse.
- Pre-empt the three honesty caveats out loud and early (shadow mode, simulated feed, structurally-not-field-proven).
- Spend the most time on the Review tab — it is the scientific heart.
- When you show the Review inlet-trajectory, point at the start-up transient yourself and name it an EnergyPlus warm-up artifact.
- Always state the scope when you quote a number: 'hall-scoped HVAC energy', 'plan-predicted PUE', 'realized stand-in'.
- Memorize the safety spine cold: 26 C hard cap, $k=1$ sigma pre-tighten, +1-sigma hot week, CVaR_{0.8}, CHWST-first ladder, zero-tolerance backstop, realized peak 25.60 C, margin 0.40 C, zero violations.
- Click live sparingly and only where it adds proof: change the plan selector on the 3D or Review tab once to show every panel re-skins from one recommendation.json (one consistent object), and let the Live heat-map sit so the green all-clear reads.
- Use the staircase narrative on History: 0% -> 3.2% -> 3.6% is the gate releasing savings as confidence grew, not the optimizer ramping up.
- On New Plan, make the point that safety is not a knob — there is no 26 C field on that form by design.
- Keep a one-slide backup of the airflow=15.5% / CHWST=3.4% / SAT=0.5% sensitivity sweep.